

AI TrustScore

Explainable trust scoring for online shopping

Problem

Fake reviews, unreliable sellers and misleading prices — shoppers can't check it all before buying.

What it does

A browser extension that gives any product page a 0–100 trust score, a risk level (Low / Medium / High) and short, plain-language reasons.

Browser extension

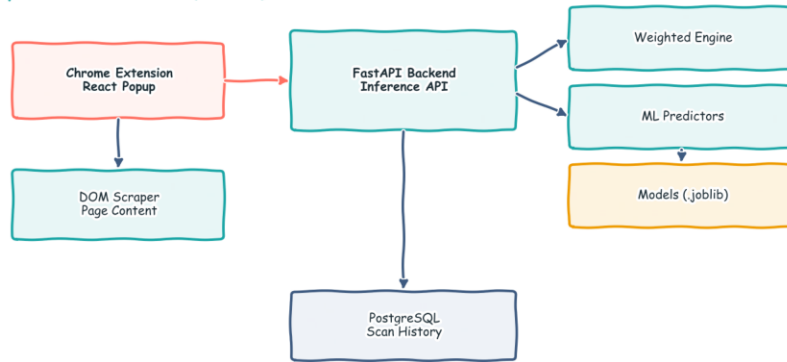
FastAPI backend

5 ML signals

Iterative Stages

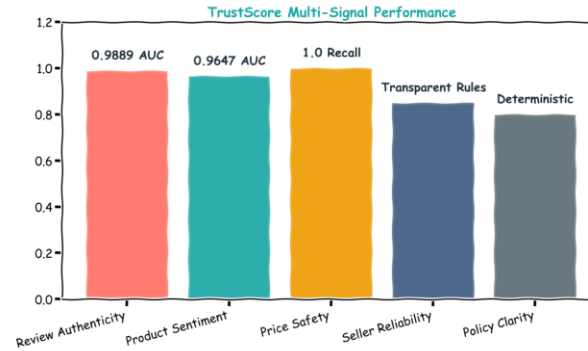
System Design & Implementation

System Architecture (Sketch)



- ▶ **Signals:** Authenticity · Sentiment · Seller · Price · Policy.
- ▶ **Architecture:** React Popup Extension + FastAPI (Python 3.13) backend.
- ▶ **Flow:** Real-time DOM scraping → ML inference → Explainable TrustScore.

Multi-Signal Performance Summary



- ▶ **Holistic Validation** Each signal is measured by the most relevant metric.
- ▶ **High Power** Both text-based models (Fake detection and Sentiment) exceed 0.96 ROC-AUC.
- ▶ **Robustness** Non-ML signals use deterministic rules to avoid bias and leakage.

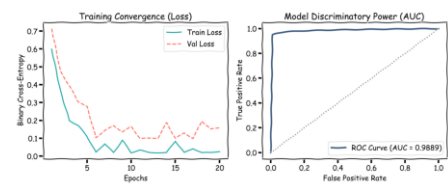
Dataset Overview & Results

Primary Training Datasets

Dataset	Rows	Use
Fake-Reviews (ArijitDas)	40,526	Fake detection
Amazon Reviews 2023	450,000	Sentiment
Amazon Metadata	180,000	Risk/Price
Yelp/IMDB/SST-2	192,349	OOD Testing

Model Components

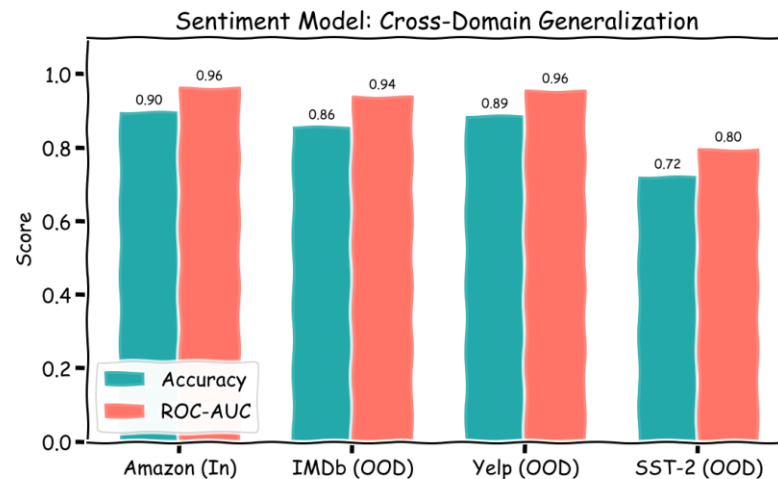
Component	Final Metric
Fake Review Authenticity	0.9889 AUC
Product Sentiment	0.9647 AUC
Seller/Price Risk	Robust Ruleset
System Calibration	ECE 0.007



Scientific Rigor & Honest Validation

Finding & Fixing Leakage

- ▶ **The Discovery:** Experimental risk models hit 1.0 accuracy. We found labels were mathematically derived from input features (leakage).
- ▶ **The Fix:** Rejected the biased model. Switched to unsupervised IsolationForest for price anomalies and transparent heuristics.
- ▶ **Validation:** Confirmed cross-domain generalization (e.g. IMDb/Yelp) to reflect real-world performance.

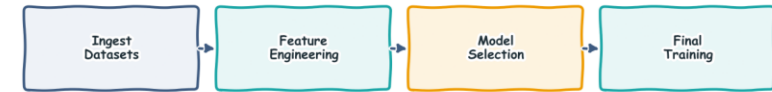


Honest domain shift validation

Conclusions & Future Work

- ▶ **Success:** Fully functional, reproducible end-to-end trust scoring system.
- ▶ **Lesson:** Scientific honesty beats hype—leakage detection led to robustness.
- ▶ **Future:** Human-labeled ground truth and DistilBERT deployment.

ML Training Workflow (Sketch)



What?

Explainable shopping trust score

Implemented?

5-signal ML pipeline + rigorous validation stages

Result?

0.9889 AUC; leakage found & fixed; calibrated probabilities

Develop further?

Real labels, transformer fine-tuning, public release